

What is linguistics?

Unit 1, Class 6

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Linguistics

- Scientific study of human language
- Not just the study of one language, but human *language* broadly
 - Linguists study individual languages as a tool to answer larger questions

Questions in linguistics

- Are there any linguistic universals? Why/why not?
- How many languages are spoken in Nepal/globally/Kathmandu?
- What determines the critical age of language acquisition? Can it be extended?
- When/why do people misunderstand written/spoken utterances?
- How can a robot sound more humane-like?
- Where does meaning arise from?
- Does speaking *Munda* or *Nepali* or *Tamang* change the way you view the world?
- How is second language acquired? How does it help in acquiring the third language?
- Will an alien have a similar language (grammar, meaning) like ours?

Descriptive vs Prescriptive

Descriptive

- Linguistics *describes* what is and was.
- Rules follow what people say.
- Concerned with linguistic knowledge.

Prescriptive

- Some forms deemed inappropriate or wrong.
- People should follow what the *rules* say.
- Current usage is generally condemned.

Methods in linguistics

- Fieldwork
- Formal analysis of patterns in data sets
- Psycholinguistic experiments
- Computational modelling
- Corpus analysis

Phonetics

Study of speech sounds (production and perception) as physical entities

- Aspiration: $[t]$ vs $[t^h]$
- Voicing: $[b]$ vs $[b^h]$

$[v]$ is pronounced by

- bringing the lower lip into contact with upper teeth and
- forcing air out of the mouth while
- the vocal folds vibrate and
- nasal cavity is closed off

Phonology

Study of sound system/patterns of a particular language, language sounds as abstract entities

- English: a word never starts with $[kn]$ while in German it is possible (e.g., *Knabe*='boy')

Morphology

Study of word structure and systematic relations between words.

- missile: 'ICBM'
- anti-tank-missile: 'missile targetting tanks'
- anti-aircraft-missile: 'missile targetting aircraft'
- anti-missile-missile: 'missile targetting ICBMs'

Morphological Rules

Rule: Anti-X-missile is a missile targetting Xs.

- What kind of missile targets anti-missile-missiles?
- anti-anti-missile-missile-missile
- anti-anti-anti-missile-missile-missile-missile: 'missile targetting anti-anti-missile-missile-missiles'

Syntax

Study of phrase and sentence structure

- word order: The book is on the table vs Table is on the book.
 - agreement: I am here vs I are here.
 - cases: I like her vs I like she.
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- I [saw [the woman] [with the telescope]].
 - I [saw [[the woman] [with the telescope]]].

Semantics

Study of linguistic meaning, i.e., meaning of sentences, phrases, words and morphemes

- What is the meaning of “I shot an elephat in a Navy with a Canon”?

Syntactically well-formed, semantically nonsensical: *Colorless green ideas sleep furiously.*

Pragmatics

Study of language use

- Context influences the interpretation of utterances – the same sentence can have different meanings across different situations.

Simply put:

- semantics: *literal* meaning and
- pragmatics: *intended* meaning.

Linguistics beyond grammar

Historical Linguistics

- How languages change over time.

Sociolinguistics

- How languages vary socially.
- How language is used as a social resource.

Psycholinguistics

- How language is represented and processed in people's mind and brain.
- Psychological and neurobiological factors associated with language acquisition and use (production, comprehension).

Language Acquisition

- How people learn language (e.g., first language acquisition).

Computational Linguistics

- Making computers process (generate/'understand'/translate...) human languages.
- Mathematical representation of natural language to understand the properties of language and how humans process language

References

Eckert, P., & Sag, I. A. (2011). Linguist 1: Introduction to Linguistics. Stanford University.

Hana, J. (2020). Introduction to linguistics: Basic concepts of linguistics [Slide deck]. Institute of Formal and Applied Linguistics (ÚFAL), Charles University.